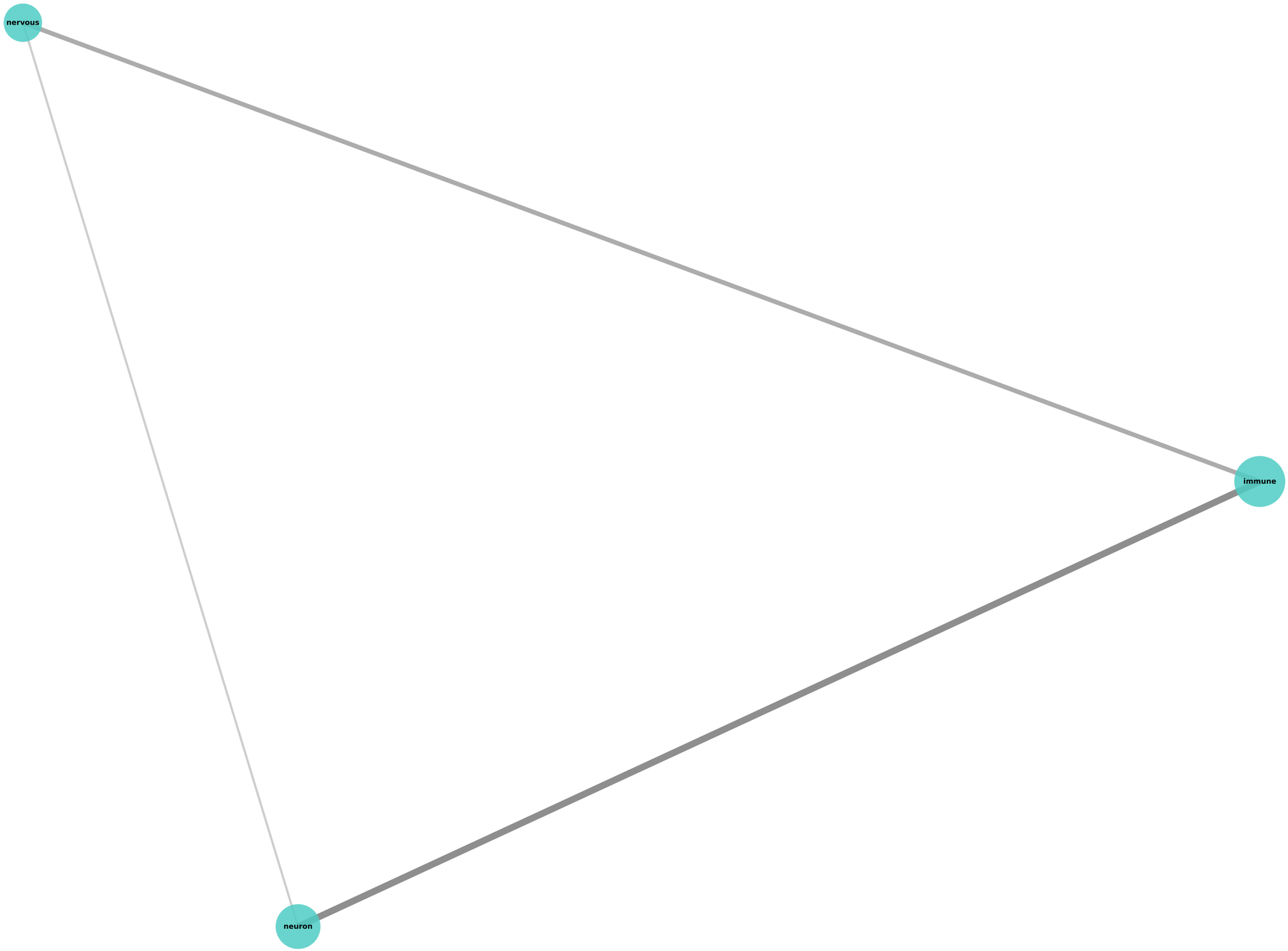Category group connectivity — LR edges